"Prevalence of undernourishment of learners from K-6 under Division of City of Sto Tomas."

A Data Engineer Pathway Capstone Project

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Chapter 1: Introduction

The City of Sto. Tomas was created by the Republic Act 11086 and got the cityhood last September 7, 2019. In this Republic Act article 8 section 48 - the creation of a city schools division and maintained by the Department of Education (DepEd) a city schools division of the City of Sto. Tomas whose area of jurisdiction will cover all the school districts within the City. Thus the creation of the Schools Division of Sto. Tomas was founded in 2022 and its Division Superintendent Dr. Neil Angeles, CESO

School feeding programs are beneficial for the physical, mental, and psychosocial development of school-age children and adolescents, particularly those in low- and middle-income countries (LMICs). While school feeding programs are ubiquitous in LMIC s, the specific benefits of school feeding programs are unclear. The aim of this systematic review and meta-analysis is to evaluate the impacts of school feeding programs on the educational and health outcomes of children and adolescents in LMICs. (Wang,2020)

The feeding programs such as milk and healthy foods must be implemented as per compliance of the Republic Act 11037 known us “Masustansiyang Pagkain para sa Batang Pilipino Act” and Department Order 38 series of 2022 of the Department of Education. And Division Memorandum number 292 series of 2023. This capstone is about the prevalence of the undernourishment of the learners from K-6 under the Division of Sto. Tomas to determine the progress of the learners from malnourished to healthy status using the database and data visualization.

Statement of the problem

1. Is the feeding program recorded in their system using MS Excel to track the progress of the learners and assessment in the weighting exam if the learners are healthy or not?

2. Can the School Division Office of Sto Tomas City comply with RA 11037 and track what are the list of schools needed to focus on the feeding program of the learners.

3. What are the timeline or best practices to reach their goals to prevent malnutrition of the learners under the Division of Sto Tomas City.

4. If there is an intervention of the local government unit of Sto. Tomas to initiate the program of feeding programs as part of the “Aksyon Bilis” program of City Mayor Arth Jhun Marasigan?

Scope and Limitations:

Scope:
1. This study is to discuss the measures and practices in feeding programs under Division of Sto. Tomas

2. To provide the relational database and normalization to generate the data reports and visualization to ensure what schools need to focus on feeding programs.

Limitations:

1. This study is limited to booster vaccination and deworming to the learners under of Division of Sto. Tomas

2. Not included in this study the assessment of literacy programs, examinations and daily Exercises (physical fitness) of the learners as part of compliance in basic education.

3. Also it is not included to make the system data analysis and visualization since the Data Engineer capstone includes the relational database and normalization.

Chapter 2: Literature Review

Educational benefits of the United States school feeding program: a critical review of the literature.

Few reports that present data from evaluations of the long-term behavioral effects of school feeding programs are reviewed. The investigations included in this section looked for effects over months or years. In contrast to the short-term effects reviewed previously, these long-term effects are more meaningful in terms of educational progress or successful adaptation to the school setting and are likely to be enduring. Lininger studied the effects of a school milk program on scholastic progress over a two-year period among 4,133 "undernourished" school students ages 6 to 16 years. The children were enrolled in special health classes in which milk drinking was one of many practices used to improve their health. The an index of scholastic progress was obtained from teachers' subjective comments. Over the period of the study, 45 per cent of the children receiving milk improved in "scholarship," whereas only 24 per cent of those not receiving milk improved. The teachers, however, presumably knew which children were and which were not receiving milk. Thus, for obvious reasons, it is impossible to determine whether the effects reported were caused by the milk supplement or by the teacher's expectations. In a year-long study, Kreitzman looked at attendance plus school grades in order to determine the effects of breakfast programs. (Pollitt,1978)

Philippine Department of Education's School-Based Feeding Program

Malnutrition is still prevalent in the Philippines. Among 12.7 million school-aged children assessed in 2019, 13.2 percent were stunted or wasted. The School-Based Feeding Program (SBFP) is one intervention to address undernutrition of school children. The Philippine Department of Education (DepEd) partners with various organizations to implement the SBFP. In 2022, the allocated budget was P3.3 Billion ($60 Million), way below the P6 Billion ($109 Million) budget allocation in 2021, as program funding shifted to local government units. The Philippine Development Plan 2023-2028 included the SBFP in the implementation of its strategy framework to improve education and lifelong learning. The proposed budget for 2023 is P5.7 Billion ($104 Million) to cover 1.6 million students. It targets stunted and wasted children.

The School-Based Feeding Program (SBFP) provides hot meals/nutritious food products (NFP) and milk to undernourished K-6 public school learners to encourage enrollment, improve classroom attendance, provide nourishment for growth and development, contribute to the improvement of their nutritional status, help boost immune systems, and enhance and improve health and nutrition values. The healthy meals provided are expected to contribute to the readiness of the learner to participate and benefit in the education process, complete their elementary education, and improve their nutritional status (Mojica-Sevilla, 2023)

A Budget and Nutrition-aware Mobile Application Featuring Food Menu Generation and Monitoring for the School Feeding Programme in Bangladesh

The school feeding program is regarded as one of the effective solutions to solve the malnutrition problem and primary school education dropout issue. Hence, to run the school feeding program effectively, a budget and nutrition-aware food menu generation scheme is required along with proper monitoring, students feedback, and real-time attendance facilities. At present, the existing related works related to

mobile application development for school feeding programs do not consider a budget and nutrition-aware food menu generation, real-time attendance system, evidence attachment of the distributed foods to ensure monitoring, students feedback regarding the program, and the donation facility at the same time. To tackle the existing challenges, this paper presents a budget and nutrition-aware mobile application featuring food menu generation and proper monitoring for the school feeding program in Bangladesh.

By using our proposed application, an authorized teacher can select the best possible food menu for students by verifying budget, nutrition, choices, and cost, and send evidence to the higher authorities regarding distributed food number, food nutrition value, and cost. Moreover, the students can give feedback to the authority regarding the school feeding program. Our evaluation results show that more than 80% of users are satisfied with the proposed system (Nath, 2022)

Impact of School-Based Feeding Program in San Luis Elementary School, Sto. Tomas South District, S. Y. 2017-2018

The results demonstrate that all the benefits under the School-Based Feeding Program at San Luis Elementary School in all 3 aspects and obtained a verbal interpretation of agreeing by all respondents. It is therefore concluded that the SBFP has a good impact on the beneficiaries. It also shows that there is no significant difference in the perception of the respondents on the benefit under the School-Based Feeding Program at San Luis Elementary School when they are grouped according to age, sex, and occupation. (Dimaano, 2019)

Table 1: Online Monitoring vs Data Engineering Approach tracker of School-based feeding program of SDO- Sto. Tomas City.

Sampling Method	Definition	Strengths	Weaknesses
Online Monitoring	In the Division Memorandum No.9 of the SDO, the feeding coordinators shall submit the online monitoring tool after class every Friday from 3:00pm onwards. Each school shall send a collage picture to their district feeding coordinators as proof to their implementation of the milk program. Also, the School Head shall oversee the daily implementation of the feeding program including procurement, actual feeding , health and nutritional assessment, disbursement and liquidation of funds, and complimentary activities.	To monitor the school-based feeding program by sending the collage picture of the learners and serves as a documentation purpose to determine if the program is implemented. Also, the coordinators and persons-in authority are involved in the activity during that event.	There is no micromanagement approach if the learner is surpassed from malnourished to healthy status and only the document serves if the program is implemented or not.

Data Engineering Approach	In the Data Engineering Approach, the author shall propose the fill-out document format. It includes the list of accessed learners, event of the feeding program, facilitator, feeding program components such as milk / food packs i.e nutribun or hot meals to lessen the process time of the accessors.	To track the list of events if the program is implemented by submitting all the necessary documents such as weighting assessment, feeding monitoring report and conduct the data visualization / analysis if the program is effective or not.	The only challenge is the cost of the software used to create the data visualization / analysis, training of non-teaching staff to create the report and must pass the compliance assessment in relation to the Data Privacy Act of 2012 such as learners name, age, sex and health status tagged as sensitive data.

Gap Analysis:

There are 4 levels of Gap Analysis in comparison of data engineering approach and online monitoring.

1. Time Sensitivity: Real-time analysis of data forms the basis of online monitoring, thereby allowing rapid decision making. On the other hand, the data engineering approach could have involved batch processing, which could introduce a delay in data availability for analysis.

2. Scope: Data engineering is all about data infrastructure and systems and it is broader as compared to online monitoring which is usually based on particular metrics or events.

3. Data Volume and Velocity: Data engineering normally deals with lots of data, and focuses on maximizing processing and storage. However, online monitoring may be suitable to handle smaller datasets but faster processing to cope with the data velocity.

4. Proactive vs Reactive: Online monitoring is more likely to target the detection of issues or opportunities on-the-go. Data engineering, although forward-looking with respect to data provision and quality, could include some more reactive tasks like trouble-shooting or system maintenance. (techtarget.com)

Chapter 3: Proposed Research Design

Step by Steps Research Design to Solve the Problem

1. Combination of OLTP (Online Transaction Processing) and OLAP (Online Analytical Processing). In the data engineering approach defined by Amazon Web Services, OLAP is optimized for complex data analysis and reporting, while OLTP is optimized for transactional processing and real-time updates. In this capstone, OLTP approach is used for the facilitators who conduct the feeding program needs to update the information of the learners, timeline of the feeding program and what are the components of the feeding program i.e milk, nutribuns and hot foods. While the OLAP is for the proposed data analytics team under SDO-Sto Tomas City who cleans the data, creates the reports and visualization, is able to submit to the Feeding Program committee to assess if the feeding program is compliant in accordance with RA 11037.

2. In this study, using the relational database such as ERD (Entity Relationship Diagram) and Normalization is able to develop clean data, showing the relationships among all available data.

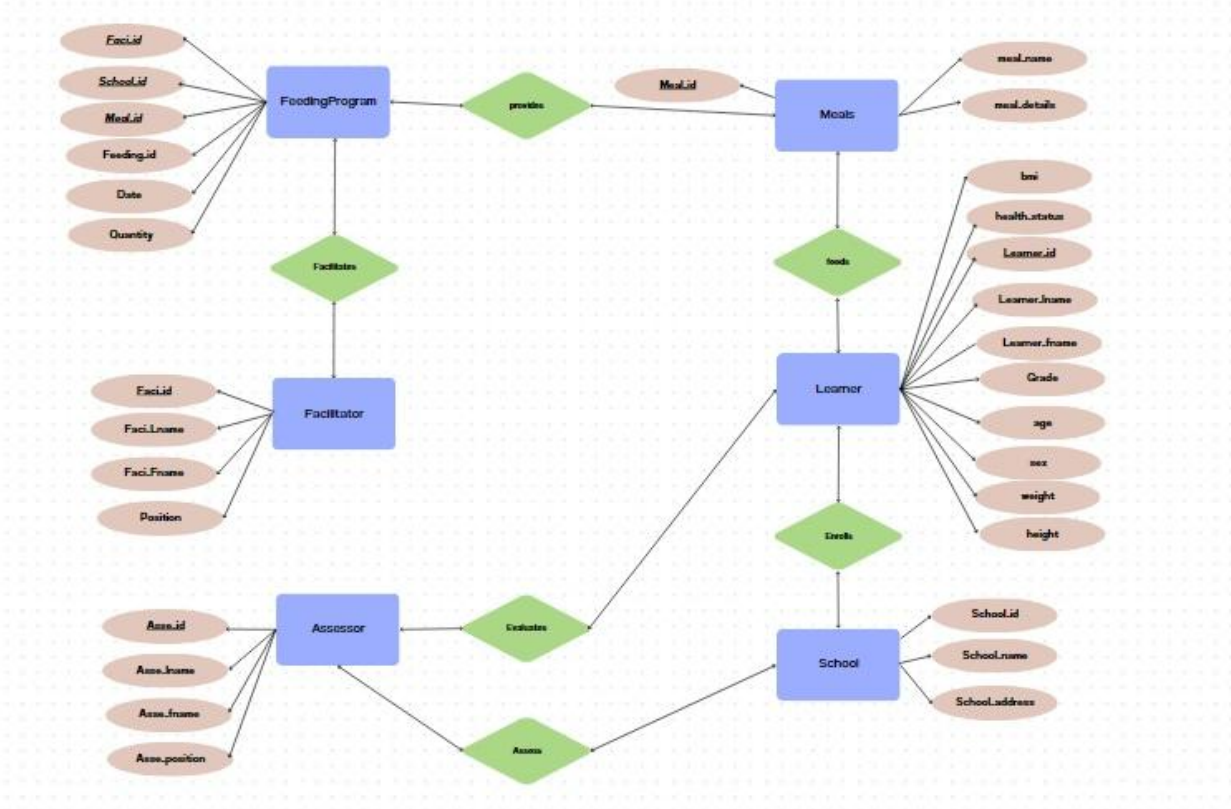
3. Creation of the proposed dashboard (visualize only) to show the reports should be done by the data analytics team to track the progress of the learners whether it is a healthy or malnourished status.

4. There is a proposed database format of all the learners and their health status to determine if they are healthy or not.

5. Using the MS Excel as the sample dataset formatted to lessen the work of the analytics team to deliver the reports in a precise way

Chapter 4: Normalization / Entity Relationship Diagram

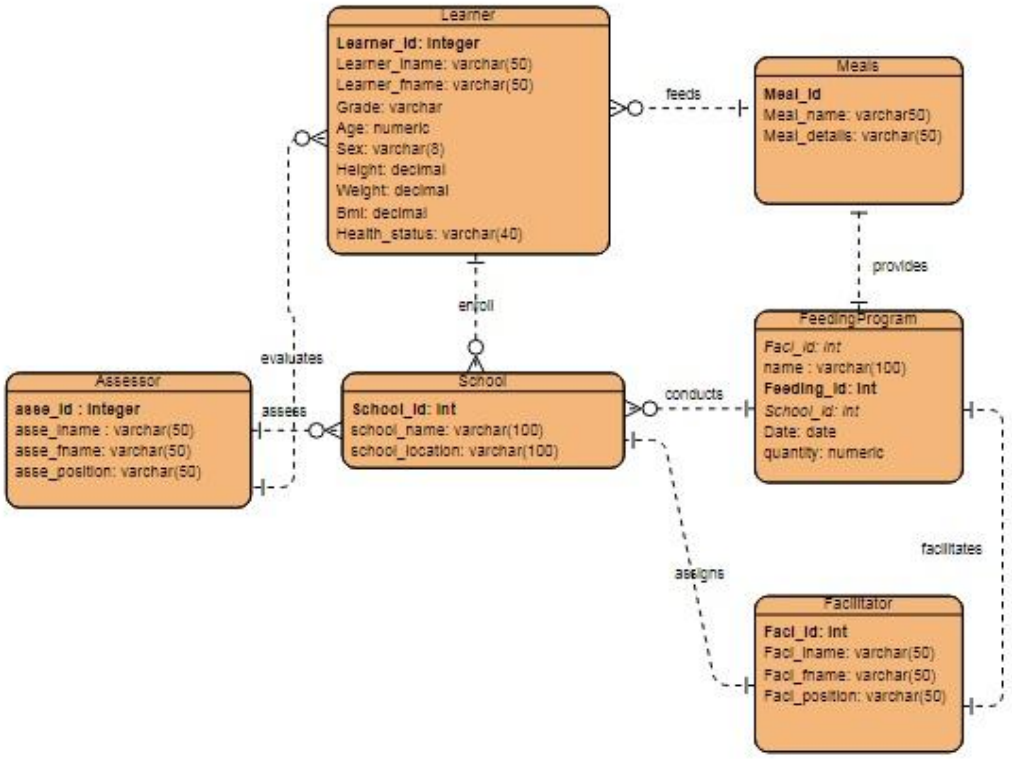
Picture 1: Entity Relationship Diagram



In this picture, it shows the function of each entity such as

a. Feeding Program - the attributes are, Feeding_id serves as a primary key. The foreign key of these attributes are School_id, Meal_id, Faci_id. The other attributes are date on the feeding program implemented and quantity of food they facilitate.
b. Facilitator - it contains, Faci_id as a primary key, faci_lname, faci_fname, and the faci_position of the facilitator who facilitates the event.
c. Assessor - these are asse_id as a primary key, asse_lname, asse_fname and asse_position who assess if the feeding program is implemented or not. The facilitator and the assessor is not the same person at the same school; it should be from the representative of the Schools Division Office.
d. Meals - it includes meal_id as a primary key, meal_name, and meal_details. For example: Meal_id : 120523001, nutribun, bread
e. Learner - Learner_id is the primary key, it includes Learner_lname, Learner_fname, grade, age,sex, weight, height, health_status and bmi serves as an information about the learner.
f. School - school_id is the primary key, the school information such as school_name and school_address is an important detail as well.

Picture 2: Entity Relationship Diagram in Cardinality



As in illustration above, here are the highlights:

a. The school enrolls many learners and the learners have dedicated school only.
b. The facilitator assists the school on the feeding program. The school also has one facilitator on the feeding program such as TLE teacher.
c. The school conducts multiple feeding programs depending on the food supply that is distributed to the learners. Also the feeding program is conducted in schools on multiple events such as the milk program, bread [nutribun] sponsored by the private companies or local government unit.
d. Meals are also distributed to the learners and the learner feeds one meal only.
e. The assessor assesses the learner through weight / height and checks if they are malnourished or healthy.
f. The assessor tracks the school to check the number of learners with regards to their health status.

Picture 3: Normalization

First Normal Form:

In the first normal form of the school based feeding program and health assessment of the learners we assure that there are no repeating groups or arrays of data.

Facilitator	
Faci_id	int
Faci_Lname	varchar
Faci_Fname	varchar
Faci_position	varchar

Learner	
Learner_id	int
Learner_lname	varchar
Learner_fname	varchar
Learner_grade	varchar
Learner_age	numeric
Learner_sex	numeric
Learner_height	decimal
Learner_weight	decimal
BMI	decimal
Health_status	varchar

Assessor	
Assess_id	int
Assess_Lname	varchar
Assess_Fname	varchar
Assess_Position	varchar

Meals	
Meal_id	int
Meal_name	varchar
Meal_details	varchar

FeedingProgram	
Faci_id	int

name	varchar
Feeding_id	int
School_id	int
Date	varchar
Quantity	numeric

School	
School_id	int
School_name	varchar
School_address	varchar

Example:
Facilitator

Faci_Id	Faci_Lname	Faci_Fname	Faci_Position
93453	Swift	Taylor	TLE Coordinator
10923	Mendoza	Lailani	TLE Coordinator

School

School_id	School_Name	School_Address
100025	Sta. Clara ES	Sta. Clara, Sto. Tomas City
100028	San Francisco ES	San Francisco, Sto Tomas City

Second Normal Form:
In the second normal form, creation of the table Feeding Program to see the feeding event id, name of the event and the date happened. Also, in the assessment to the learners conducted by the assessor, separation of the table for assessment for learners and to school to ensure we complied to the 2nd normal form

FeedingProgram	
Feeding_id	int
Name	varchar
Date	varchar

LearnerAssessment	
Assessment_id	int

Learner_id	int
Assessment_date	varchar
Weight	decimal
Height	decimal
BMI	decimal
Health_status	varchar

SchoolAssessment	
Assessment_id	int
School_Id	int
Number_of_Learners	numeric
School_Health_Status	varchar

Example:
LearnerAssessment

Assessment_id	Learner_id	Assessment_date	Weight	Height	BMI	Health_status
89233	2020120	12-01-23	20	100	22	Healthy

School_Assessment

Assessment_id	School_id	Number_of_Learners	School_Health_Status
89233	100025	700	Good

Third Normal Form
In the third normal form, the Learner are stored in a separate table to ensure atomicity. Also the learner table has the foreign key referencing the school table. Each school has the facilitator and assessor has assessed random schools to ensure the data integrity.

Learner	
Learner_id	int
Learner_Lname	int
Learner_Fname	varchar
Grade	varchar
School_Id	int

FacilitatorSchools	
Faci_id	int
School_id	int

AssessorSchools	
Assess_id	int
School_id	int
Assessment_id	int

Example:

Learner_id	**Learner_Lname**	**Learner_Fname**	**Grade**	***School_id***
2020120	Bernardo	Kathryn	4	100025

Chapter 5: Conclusions and Recommendations

The Schools Division Office of the City of Sto Tomas was newly created last 2022 and upon doing this capstone project. The researcher coordinates with the officials and health professional staff of the Schools Division Office to provide the dataset since there are pending documents and needs to be found in the City Health Office and there is no guarantee time when they are provided.

The statement of the problem provided in the chapter 1 is not answerable at this time since the SDO is newly created and its run on its first operation and still organized to provide the best service to the learners.

Also, if the SDO- Sto Tomas City wants to provide the data visualization and analysis. The researcher will continue this research when the researcher gets an approval to enroll to the Data Scientist Pathway from the Project SPARTA team next year. Besides, the Data Engineering Pathway Capstone has to create the Normalization and Relational Database.

This is the initiative to provide the ideas and best practices in the data community to improve the better lives of the Filipinos through adoption. Not only this research but to provide data availability for the next researchers.

This information on the table provided is dummy in compliance with the Data Privacy Act of 2012 since this is the proposed research only.

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